

Impact of the Mid Day Meal Scheme on Primary School Education in Tamil Nadu

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B. A. Multidisciplinary (Economics + Political Science)

Introduction

The Nutritious Meal Programme was introduced in Tamil Nadu by the then Chief Minister Puratchi Thalaivar Dr. M.G.R. in July 1982. The underlying objective was to emphasise education along with nutrition, wherein hot, nutritious meals were to be served to children in government schools, government aided schools, special training centres and madarsas. These initiatives have continued to this date, achieving the aims of maximising enrollment, reducing dropout rates, and providing nutrition to underprivileged children. The idea was to “universalise elementary education” (Government of Tamil Nadu). It was initially introduced for rural children in the pre-school age group of 2-5 years, and primary school children in the age group 5-9 years (Swaminathan et al., 2004). From September 1982, the scheme was extended to the urban area, included old-age pensioners from 1983, and brought children in the age group 10-15 in its ambit from 1984 (Swaminathan et al., 2004).

“Tamil Nadu’s noon-cum-nutritious meal scheme is the country’s largest in terms of the number of beneficiaries covered” (Swaminathan et al., 2004). This scheme was implemented nationwide 1995 onwards in the name Mid Day Meal Scheme by the intervention of the Supreme Court. It was a welcome initiative as nutritional deprivation in children often leads to “educational as well as physical underdevelopment” (Padhi, 2009). Additionally, the poor nutritional status of the school going children is a significant cause of dropping out, slow learning and failure to learn in primary stages. The MDM scheme was seen as a way to “to improve the health and nutritional status of children and to develop their mental and physical ability” (Policy Note, Government of Tamil Nadu, 2002-2003).

Literature Review

As of the 2022-23 calendar, the total number of schoolchildren in Tamil Nadu was 1.29 crores approximately (Education for All in India Report, 2026). The MDM Scheme has ensured “complete servicing of the child and the mother from the time of its creation in its womb” (Arumugam, 2016). Way back in 1925, a midday meal programme was introduced for children belonging to poor socio-economic strata in the Madras Corporation. Over the years, the Government of Tamil Nadu experimented with different schemes to enhance enrollment and

nutritional status of the children, and curtail dropouts. The Mid Day Meal Scheme was one of such schemes. There exist various justifications for rolling out this scheme namely “political, nutritional, employment, welfarist, health, education and social” (Swaminathan et al., 2004). It has been observed that the noon-meal scheme helps foster a spirit of community within young children, as children across genders, social stratas, castes, and classes eat and share food (Arumugam, 2016). There has been evidence of falling dropout rates in Tamil Nadu after the implementation of the noon-meal scheme, while the remaining rates are assumed to be sustained by inability to cope or disinterest in studies (Swaminathan et al., 2004).

A study conducted in Virudhunagar district highlights the role of the Mid-Day Meal Scheme (MDMS) in improving school participation, especially among children from economically weaker backgrounds (Dhatchayani et al., 2024). The study finds that there is a clear relationship between the occupation of parents or guardians and school attendance, indicating that children from underprivileged households benefit significantly from such welfare schemes. In this context, the MDMS helps in reducing inequalities and promoting access to education (Dhatchayani et al., 2024).

The findings also show that a large majority of students (around 85%) believe that the scheme has increased their attendance in school. This suggests that the provision of meals acts as an incentive for regular attendance. Additionally, about 78.33% of respondents agreed that the scheme encourages children from poorer families to attend school instead of engaging in work, pointing towards its role in reducing child labour (Dhatchayani et al., 2024). However, the scheme has also been criticised on grounds of “being a drain on government resources” and “diverting resources that could have otherwise been used for 'hard' development activities” (Swaminathan et al., 2004).

There also exist various problems within the scheme. The list of beneficiaries is inflated and overstated in almost all the districts, and this “translates into larger quantities of ration than actually required” (Swaminathan et al., 2004). However, this is offset by “supply of less than stipulated quantity of ration” while the officer mentions full access in the records, even when not applicable. Additionally, there are leakages of the ration at the local level where the “per capita allocation per child is also reduced by the organisers and cook” (Swaminathan et al., 2004). Concerns have also been raised regarding the poor quality of the ration and the substandard conditions of the kitchen and the utensils (Swaminathan et al., 2004).

Methodology

To analyse the impact of the Mid Day Meal Scheme (now Pradhan Mantri Poshan Shakti Nirman Yojana) on primary school education in Tamil Nadu, this paper undertakes an econometric analysis of time series data from 2003-2004 to 2024-2025. This has been done by running an

OLS Regression. The dataset includes both raw and transformed variables for regression analysis.

- Independent Variable** in the study is the annual budget allocation for the MDM scheme at the national level, collected from sources including GOI Reports, HRD Reports, PM POSHAN website and Press Releases. The national data has been utilised due to the unavailability of data at the state level, and additionally due to the 75:25 funding ratio for the scheme, where around 75% of the funding is provided by the Central Government. While state-level expenditure data would have been more appropriate, national level budget allocation is used as a proxy due to data limitations. This is a limitation in capturing the state specific results.
- The Dependent Variable** is the Elementary Gross Enrollment Rate (Classes I to V) in Tamil Nadu from 2003-2004 to 2024-2025, collated from a wide range of sources including Ministry of Education statistics and Kailash Satyarthi Foundation Reports. The dependent variable ideally suited to capture the impact of the Mid-Day Meal (MDM) Scheme is the primary/elementary dropout rate, as the scheme primarily affects retention rather than initial enrollment. However, due to the lack of a consistent and comparable time-series dataset for dropout rates for Tamil Nadu over the study period (2003–2025), this study uses the Elementary Gross Enrollment Ratio (GER) as a proxy indicator. It is acknowledged that GER may not fully capture the impact of the scheme, especially after the achievement of near-universal enrollment in Tamil Nadu.

Years	Budget Allocation for MDM (India) (in Crores)	Elementary GER (Tamil Nadu) (in Percentage)	Ln(BAMDM)	Ln(EGER)
2003-2004	1375	116.51	7.22620901	4.75798
2004-2005	3057	118.41	8.025189322	4.77415
2005-2006	3345.26	120.07	8.115299697	4.78807
2006-2007	5348	117.81	8.584477938	4.76907
2007-2008	6678	116.14	8.806573821	4.7548
2008-2009	8000	115.1	8.987196821	4.7458
2009-2010	7359.19	114.7	8.903705151	4.74232
2010-2011	9440	111.8	9.152711259	4.71671
2011-2012	10380	115.7	9.247636157	4.751
2012-2013	11937	115.8	9.387398099	4.75186
2013-2014	13215	114.7	9.489107827	4.74232
2014-2015	13215	110.1	9.489107827	4.70139
2015-2016	9236	104.9	9.13086417	4.65301
2016-2017	9700	103.2	9.179881164	4.63667
2017-2018	10000	102.5	9.210340372	4.62986
2018-2019	10500	100.3	9.259130536	4.60817
2019-2020	11000	99.5	9.305650552	4.60016
2020-2021	12900	100.1	9.46498259	4.60617
2021-2022	11500	99	9.350102314	4.59512
2022-2023	10233.75	95.9	9.233446361	4.56331
2023-2024	11600	95.8	9.358760377	4.56226
2024-2025	12467.39	91.6	9.430871714	4.51743

To improve model accuracy and address non-linearity, the following variables were log-transformed:

- $\ln(\text{Elementary GER})$
- $\ln(\text{Budget Allocation for MDM})$

These log transformations help interpret the results in terms of percentage changes and enhance model stability (Log-Log Model).

Microsoft Excel's Data Analysis ToolPak was used to conduct the regression analysis.

Therefore, the final regression model came out to be:

$$\ln(EGER) = \beta_0 + \beta_1 \ln(BAMDM) + \varepsilon$$

Where:

- $\ln(EGER)$ = Log of Elementary Gross Enrollment Rate (Dependent Variable)
- $\ln(BAMDM)$ = Log of Budget Allocation for MDM Scheme (Independent Variable)
- β_0, β_1 = Regression coefficients
- ε = Error term

Statistical Outcome

Regression Summary

Regression Statistics	Values
Multiple R	0.561140358
R Squared	0.314878501
Adjusted R Square	0.280622426
Standard Error	0.071077578
Observations	22

Regression Coefficients

Variable	Coefficient	Standard Error	t-Statistic	p-Value	95% Confidence Interval
Intercept	5.427084017	0.246766055	21.99283045	1.74863E-15	(4.912339046, 5.941828989)
Ln (BAMDM)	-0.082829116	0.027319977	-3.031815051	0.006586299	(-0.139817589, -0.025840644)

This gives us the regression equation:

$$\ln(EGER) = 5.4270 - 0.0828\ln(BAMDM)$$

Results and Discussion

To understand the impact of the Mid Day Meal Scheme on Elementary Gross Enrollment Rate in Tamil Nadu, we utilised a double-log model for the time-series data (2003-2025). The regression analysis yielded the results mentioned in the methodology section.

Interpretation of the Constant Term (β_0)

When the log of the independent variable, that is the Budget Allocation for the Mid Day Meal Scheme (BAMDM) is 0, then on an average, the log of the dependent variable, that is the Elementary Gross Enrollment Rate (EGER) is 5.43 (approximately). The Standard Error (SE) is 0.246, which is relatively small, indicating a reliable estimate within the dataset.

Relationship between Elementary Gross Enrollment Rate (EGER) and Budget Allocation for the Mid Day Meal Scheme (BAMDM)

The coefficient β_1 indicates that as the Budget Allocation for the Mid Day Meal Scheme (BAMDM) increases by 1%, Elementary Gross Enrollment Rate (EGER) fall by 0.0828% on an average, holding other independent variables constant. This is in contrast to our expectations. We hypothesized the coefficient β_1 to be positive, according to our estimations that as there is a rise in the budget allocation for MDM, there will be greater spending on diversification and strengthening of the nutritional infrastructure, leading to a greater access to food and thus resulting in an uptick in the gross enrollment rates. However, the regression results postulated that the relationship between them is in fact negative. Therefore, we have utilised the two-tailed hypothesis test for this relationship, with the level of significance at 5 %.

$$H_0: \beta_1 = 0$$

Ha: $\beta_1 \neq 0$

As the calculated t-statistic value -3.031 is greater than the tabulated t-statistic value of -2.086 at the 5% significance level, thus we reject the null hypothesis. There exists a statistically significant relationship between BAMDM and EGER. Additionally, the p-value here is 0.0065, which is less than the 0.05 significance level. Thus, this will also lead to the same conclusion. The SE is 0.027, indicating a reliable estimate.

This negative relationship may be explained by the saturation of enrollment levels in Tamil Nadu. Since the state had already achieved near-universal enrollment, further increases in MDM budget allocation do not translate into higher enrollment. Additionally, improvements in data systems and declining child population may also contribute to the observed decline in GER

Interpretation of R² (Coefficient of Determination)

R² showcases the percentage of the variation in the Dependent Variable (EGER) that can be explained by the regression line (relationship between EGER and the independent variable BAMDM). Here, it displays that 31.48 % variation in the gross enrollment rate is explained by the budget allocation for MDM scheme. This suggests a modest explanatory power of the model. The relatively low R² can be attributed to the fact that enrolment in Tamil Nadu has already reached saturation levels, and is influenced by several other factors such as parallel policy interventions, which are not included in the model.

Interpretation of Adjusted R²

Adjusted R² accounts for the added independent variables, and the changes in Degrees of Freedom. Here, the value is 28.06%, denoting that 28.06% of the variation in EGER is caused by the independent variable BAMDM, after adjusting for the number of explanatory variables. This suggests that a strong model fit cannot be observed.

Conclusion

This study explores the relationship between Mid-Day Meal (MDM) Scheme budget allocation and elementary Gross Enrollment Ratio (GER) in Tamil Nadu over the period 2003-2025. The analysis finds that while MDM budget expenditure has increased steadily over time, its statistical association with enrollment remains weak, as can be seen in the relatively low R² value. The estimated negative coefficient further suggests that increases in budget allocation are not associated with higher enrollment levels in the state.

However, these findings must be interpreted with caution. Tamil Nadu had already achieved near-universal elementary enrollment by the early 2010s, leaving limited scope for further improvements in GER. As a result, the MDM scheme's impact is not adequately captured through enrollment indicators. Instead, its contribution is likely reflected in improved retention, attendance, and nutritional outcomes, which are not fully captured in this analysis.

The study is also subject to certain limitations, including the use of GER as a proxy variable due to the lack of consistent dropout data, and the exclusion of other important determinants of schooling such as income levels and parallel policy interventions. These factors may lead to omitted variable bias and affect the reliability of the results.

Despite these limitations, the paper highlights an important policy insight that welfare interventions such as the MDM scheme may have diminishing marginal effects on enrollment once universalisation is achieved. Therefore, future policy focus should shift towards improving learning outcomes, retention, and quality of education rather than merely increasing enrollment.

In conclusion, while the Mid-Day Meal Scheme remains a crucial component of India's school welfare system, future research can improve upon this study by incorporating state-level expenditure data, and using dropout or attendance rates as dependent variables.

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